

Shilin (Edward) Zhu

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Education

Northeastern University	Sep 2025 — Jul 2027 (Expected)
Master of Science in Computer Science GPA: 3.92/4.00	Seattle, WA
South China Agricultural University	Sep 2011 — Jul 2015
Bachelor of Science in Computer Science and Technology	Guangzhou, China

Selected Projects

<u>Detypify</u>	May 2026 — Jul 2026
Open-source Typst symbol classifier	
• Contributed to an open-source pipeline recognizing 400+ Typst symbols, upgrading MobileNetV4 to MobileNetV5; stabilized CUDA/PyTorch training and ONNX export, and added macro-F1, confusion-matrix, and dataset diagnostics	
<u>Compact Language Model for Rust Code Generation</u>	Jan 2026 — Apr 2026
A 0.8B-parameter causal Transformer specialized for Rust code generation	
• Ran 1,000-step compiler-feedback reinforcement learning on 10,281 Rust examples with 16 rollouts per prompt in 14 hours on one H100, raising training compile rate from 35% to 75%	
• Engineered an end-to-end LLM pipeline spanning dataset preparation, BPE tokenizer training, staged pre- and post-training, checkpoint recovery, inference, and automated evaluation	
• Created a reproducible evaluation harness for loss, bits per byte, and generation analysis; documented the methodology and results in a technical report	
<u>Mulan</u>	Mar 2025 — May 2026
Agentic workflow platform for AI video generation	
• Developed an orchestration engine for prompts, assets, asynchronous execution, and outputs across multimodal video models including Veo, Kling, and Seedance	
• Built an LLM agent with memory, tool calling, planning, reflection, and output validation that converted creative briefs into executable video pipelines; integrated human-in-the-loop editing, telemetry, and production observability	

Work Experience

<u>ByteDance</u> - Douyin/TikTok	Aug 2020 — Aug 2023
Senior Software Engineer	Shenzhen, China
• Led 8 engineers building Jimu, an end-to-end visual workflow and rendering platform used by 50,000+ employees	
• Architected backend-for-frontend services in Node.js, Go, and Python linking page configuration, rendering, and content-moderation pipelines	
• Led development of Argus, a visual DAG editor and rules engine for ML-powered harmful-content detection and filtering across Douyin, TikTok, and other products	
<u>Tencent</u> - Social Network Group (SNG)	Nov 2018 — Aug 2020
Software Engineer	Shenzhen, China
• Built a distributed content-extraction platform with Redis, Puppeteer, Node.js, and Docker that processed ~10M records daily	
• Built Elasticsearch and Grafana dashboards for monitoring, data quality, and crawler effectiveness	
<u>Alibaba</u> - Tmall Global	Mar 2016 — Nov 2018
Software Engineer II	Hangzhou, China
• Built international e-commerce systems for Taobao Overseas, Lazada, and Tmall Global, supporting 5 country sites with localization, multiple currencies, and shared platform modules	
• Cut page load times in Southeast Asia by 60%+, from 5–11 seconds to under 2 seconds, through CDN acceleration, critical-resource extraction, and asynchronous loading	
<u>Lizhi Network Technology Co.</u>	Mar 2015 — Feb 2016
Junior Software Engineer	Guangzhou, China

Skills

- **ML & AI Systems:** PyTorch, CUDA, ONNX, LLM Training, Reinforcement Learning, AI Agents, Tool Calling, Multimodal APIs
- **Infrastructure:** Slurm, Weights & Biases, Docker, Redis, Elasticsearch, Grafana, Production Observability
- **Programming Languages:** Python, Rust, TypeScript, JavaScript, Go, Java, C++